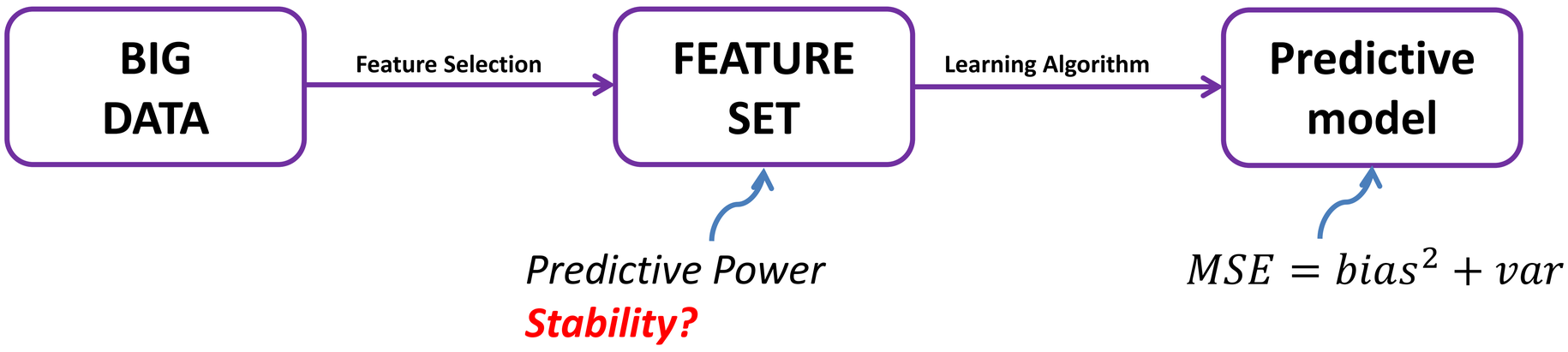
MANCHESTER 1824

The University of Manchester

Measuring the Stability of Feature Selection

Sarah Nogueira and Gavin Brown
sarah.nogueira@manchester.ac.uk



Unlike variance of prediction models, there is no unique definition of stability, with numerous proposed measures over 15 years of literature. Starting from an axiomatic point of view, we identify what properties would be desirable. Somewhat surprisingly, we found that the Pearson's correlation coefficient has all required properties.

Stability

- ***Sensitivity*** *of the feature selection (FS) procedure to* ***small perturbations*** *in the dataset*
- Extremely relevant in biomarker identification:
- Indicator of **reproducible research**
  - **A major obstacle** to clinical applications
  - In some applications, a stable set of markers is said to be **as important as** their predictive power

Example

$$A_1 = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 1\,1\,1\,0\,0 \\ 1\,1\,1\,0\,0 \\ 1\,1\,1\,0\,0 \end{bmatrix} \Bigg\} \; M = 3 \; feature\ sets$$

Fully stable, all features selected with a frequency of 0 or 1

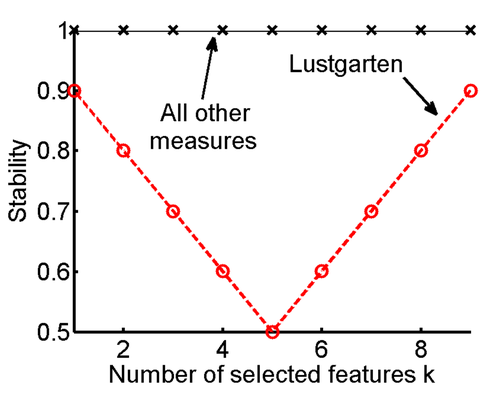
$$A_2 = \begin{bmatrix} 1\,1\,1\,0\,0 \\ 1\,0\,1\,1\,0 \\ 1\,0\,1\,0\,0 \end{bmatrix}$$

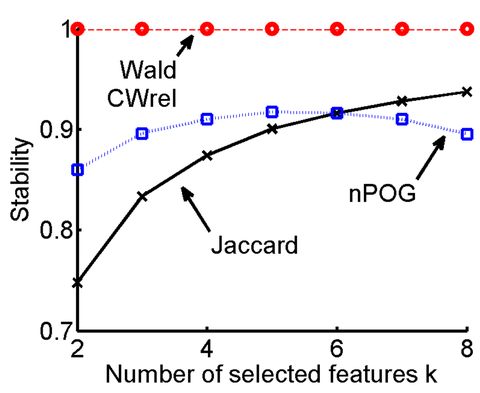
Unstable. Features 2 and 4 selected with frequency 1/3

Desired Properties

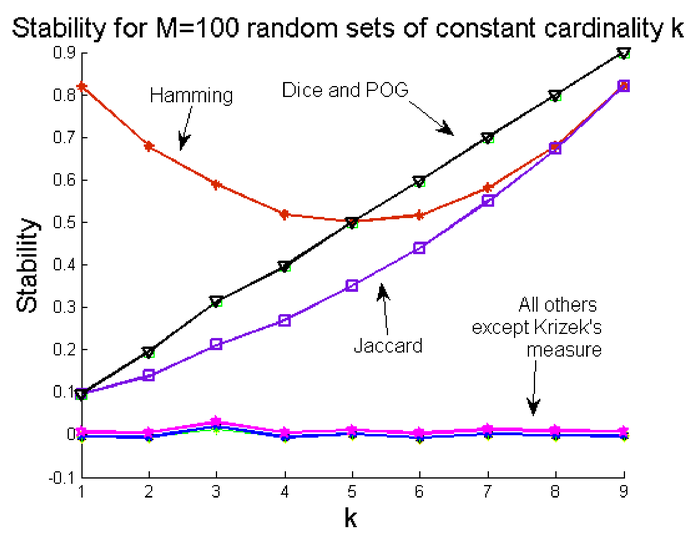
Which properties a stability measure $\hat{\Phi}$ should have so that the stability values are interpretable and comparable?

Fully defined: $\hat{\Phi}$ Defined for any sequence of feature sets
Bounds: $\hat{\Phi}$ is bounded by constants
Maximum: $\hat{\Phi}$ reaches its maximum ↔ All feature sets are identical





Correction for Chance:
$E[\hat{\Phi}] = 0$ when the selection is random

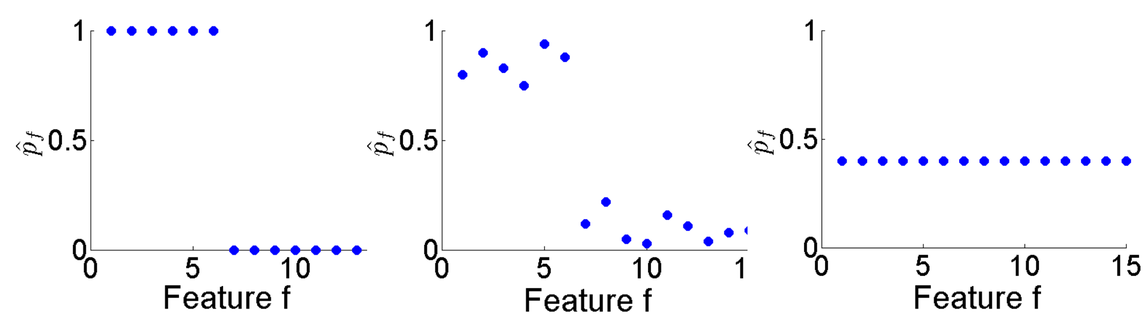


Existing Measures

	Fully defined	Bounds	Maximum	Correction For Chance
Jaccard (2002)	✓	✓	✓	
Hamming (2007)	✓	✓	✓	
Dice (2008)	✓	✓	✓	
POG (2006)	✓	✓	✓	
Kuncheva (2007)		✓	✓	✓
nPOG (2009)	✓		✓	✓
Lustgarten (2009)	✓	✓		✓
Wald (2013)	✓			✓
Krizek (2007)			✓	
CWrel (2010)	✓	✓		✓
Pearson	✓	✓	✓	✓

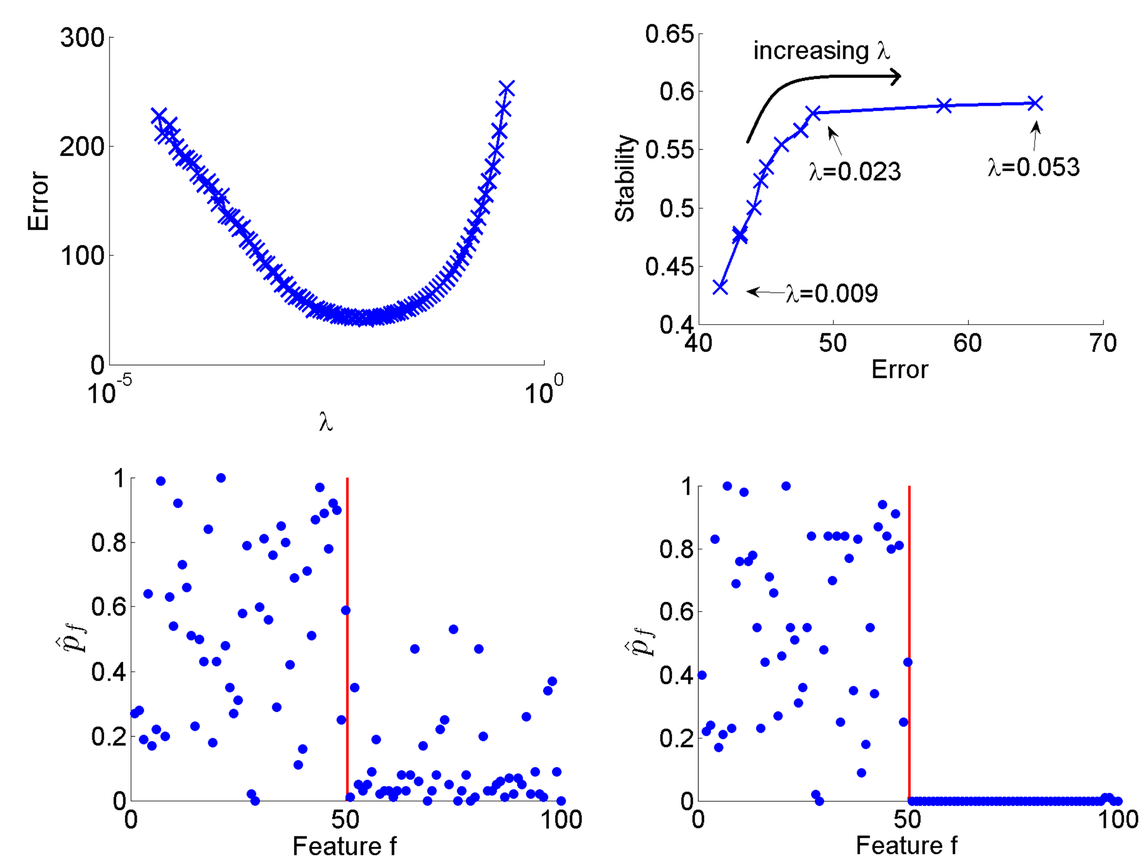
Using Pearson's correlation

- Define stability as the average pairwise correlations between all feature sets
- **Has all desired properties**
- Is a **generalization of Kuncheva's measure** for feature sets of varying cardinality



Experiments

Dataset where the first 50 features are relevant



Implementation in Matlab available online at: https://*github.com/nogueirs/ECML2016*